2.2 The Golden Ratio as a Limit

WORKSHEET

The question

STEP II 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Do you recognise the numerators and denominators? And look at the decimals: $1, 2, 1.5, 1.667, 1.6, 1.625, 1.615, \ldots$. Is the sequence increasing? Decreasing?

STAGE 1 Comparing terms two apart

Covers: Part (i)

Ask yourself Use $(*)$ to write $u_{n+2} - u_n$ as $\frac{1}{u_{n+1}} - \frac{1}{u_{n-1}}$, and put it over a common denominator.

Pause and try Do the same for $u_{n+1} - u_{n-1}$, and combine the two. Then use $(*)$ to simplify the products $u_{n+1}u_n$ and $u_{n-1}u_{n-2}$ in the denominator.

STAGE 2 A trap

Covers: Part (ii)

Pause and try Before using part (i), prove by induction that $1 \leq u_n \leq 2$ for all n. If $1 \leq u_n \leq 2$, how big can $\frac{1}{u_n}$ be?

STAGE 3 Two monotone subsequences

Covers: Part (iii)

Ask yourself Part (i) says $u_{n+2} - u_n$ is a positive multiple of $u_n - u_{n-2}$. So what can you say about the signs of $u_3 - u_1$, $u_5 - u_3$, $u_7 - u_5$, ...?

Ask yourself Do the odd terms increase or decrease? And the even terms? Are they bounded?

Pause and try The theorem you are given is about increasing sequences. How can you apply it to a decreasing sequence that is bounded below?

STAGE 4 Finding the limits

Covers: Part (iii)

Ask yourself Use $(*)$ twice to write u_{n+2} directly in terms of u_n.

Pause and try Let $n \to \infty$ through odd values, then through even values. Both limits satisfy the same quadratic: which root is it? Why does the whole sequence then converge?

STAGE 5 Starting somewhere else

Covers: the last part, starting from $u_1 = 3$

Ask yourself With $u_1 = 3$, write down the first few terms. Which of your earlier arguments still work, and from which term onwards?

What's really going on

Cobwebs and fixed points

Every sequence defined by $u_{n+1} = f(u_n)$ can be drawn as a *cobweb diagram* (Figure 2.2.1). Plot $y = f(x)$ and $y = x$. Start at u_1 on the x-axis, go vertically to the curve (height $u_2 = f(u_1)$), then horizontally to the line $y = x$ (which moves you to $x = u_2$), then vertically to the curve again, and so on. Here $f(x) = 1 + 1/x$ is decreasing, which is why the path spirals inwards, alternately overshooting and undershooting, towards the point where the curve crosses the line: the *fixed point* φ, where $f(\varphi) = \varphi$.

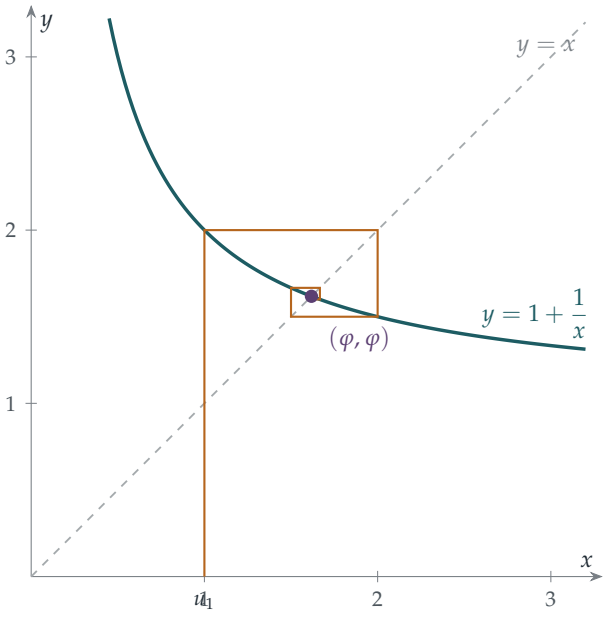


Figure 2.2.1 A cobweb diagram for $u_{n+1} = 1 + 1/u_n$ starting from $u_1 = 1$. The path goes up to the curve and across to the line $y = x$, over and over. Because the curve slopes downwards, the path spirals in towards the fixed point $\varphi \approx 1.618$, alternating sides: that is the zig-zag of odd and even terms.

How fast, and why

Where does the factor $(1 + u_n)(1 + u_{n-2})$ in part (i) come from? Near the fixed point, $u_{n+1} - \varphi = f(u_n) - f(\varphi) \approx f'(\varphi)(u_n - \varphi)$, and $f'(\varphi) = -1/\varphi^2 \approx -0.38$. So each step multiplies the error by about -0.38: the sign flips (zig-zag) and the size shrinks (convergence). Two steps multiply it by $1/\varphi^4 \approx 0.146$, which is what $1/\big((1 + u_n)(1 + u_{n-2})\big)$ tends to. At university this becomes the *contraction mapping theorem*: if $|f'| \le k < 1$ near a fixed point, iteration converges to it, with the error shrinking at least geometrically. It is one of the most useful theorems in mathematics, and it is how you prove that differential equations have solutions (Chapter 8). In Section 2.8 we will meet a map whose fixed points are *not* attracting, with very different results.

The most irrational number

The terms $u_n = F_{n+1}/F_n$ are the best possible fractional approximations to φ for their size, but "best" is not very good: φ is harder to approximate by fractions than any other number. The reason is visible in its continued fraction $[1; 1, 1, 1, \ldots]$: large entries in a continued fraction mean very good approximations (think of $\pi = [3; 7, 15, 1, 292, \ldots]$ and the excellent $\frac{355}{113}$), and φ has the smallest possible entries all the way. Adolf Hurwitz proved in 1891 that every irrational x has infinitely many fractions with $|x - p/q| < 1/(\sqrt{5}\,q^2)$, and that $\sqrt{5}$ cannot be improved, precisely because of φ. This "maximal irrationality" is the reason the golden angle appears in the spiral patterns of

sunflower seeds.

Try this next

1 **Fibonacci inside.** Prove that $u_n = F_{n+1}/F_n$, where $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$.

2 **The error shrinks.** Show that $u_{n+1} - \varphi = -\dfrac{u_n - \varphi}{\varphi\, u_n}$, and deduce that $|u_n - \varphi| \le \varphi^{-(n-1)}$ for all n. (Use $\varphi^2 = \varphi + 1$, and $u_n \ge 1$.)

3 **Breaking the sequence.** Show that if $u_1 = \frac{1-\sqrt{5}}{2}$ then the sequence is constant. Find all the starting values u_1 for which some $u_n = 0$, so that the sequence cannot continue. (*Hint:* work backwards from $u_n = 0$.)

4 **A cousin for $\sqrt{2}$.** Let $v_1 = 1$ and $v_{n+1} = 1 + \dfrac{1}{1 + v_n}$. Adapt the method of this question to show that $v_n \to \sqrt{2}$, and write $\sqrt{2}$ as a continued fraction.

Further reading

- **Mario Livio, *The Golden Ratio*** (Broadway). The history of φ, separating the genuine mathematics from the many myths.
- **C. D. Olds, *Continued Fractions*** (Mathematical Association of America). A short, very readable introduction aimed at strong school students.
- **J. C. Burkill, *A First Course in Mathematical Analysis*** (Cambridge University Press). Rigorous treatment of monotone sequences and completeness, with many exercises.

The full guided solution is in Section 2.2:
https://beyondstep.keithmcnulty.org/ch02/golden-limit.html